Database 1

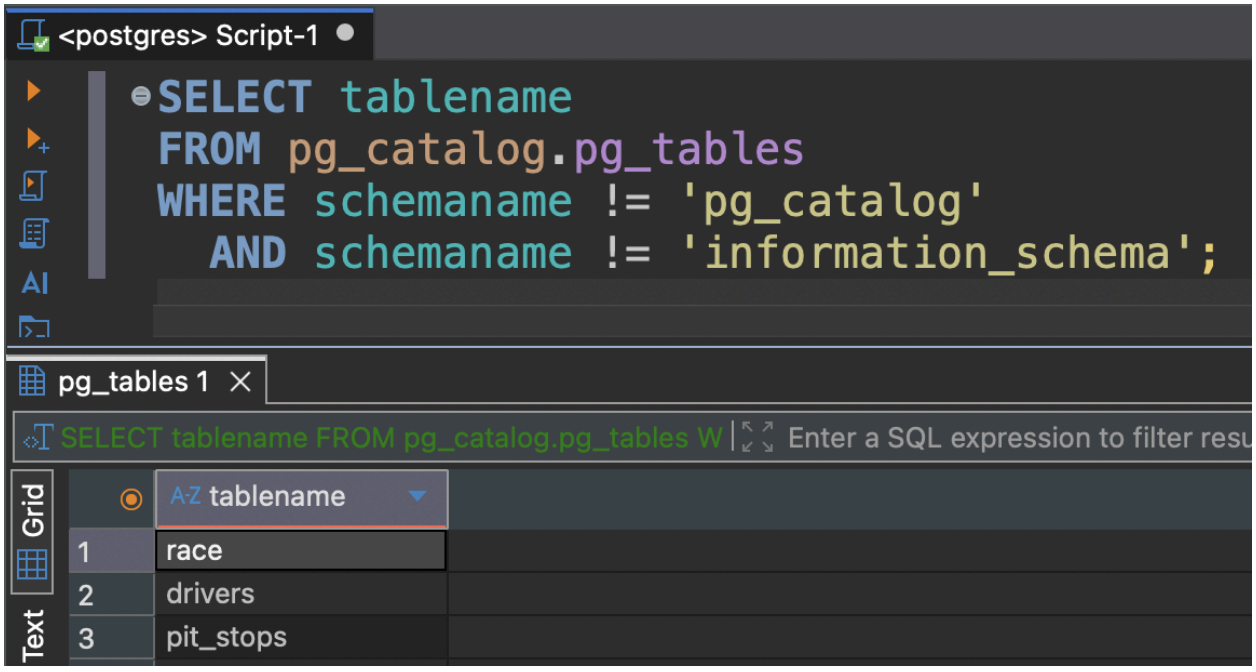
```sql
PRAGMA foreign_keys = ON;

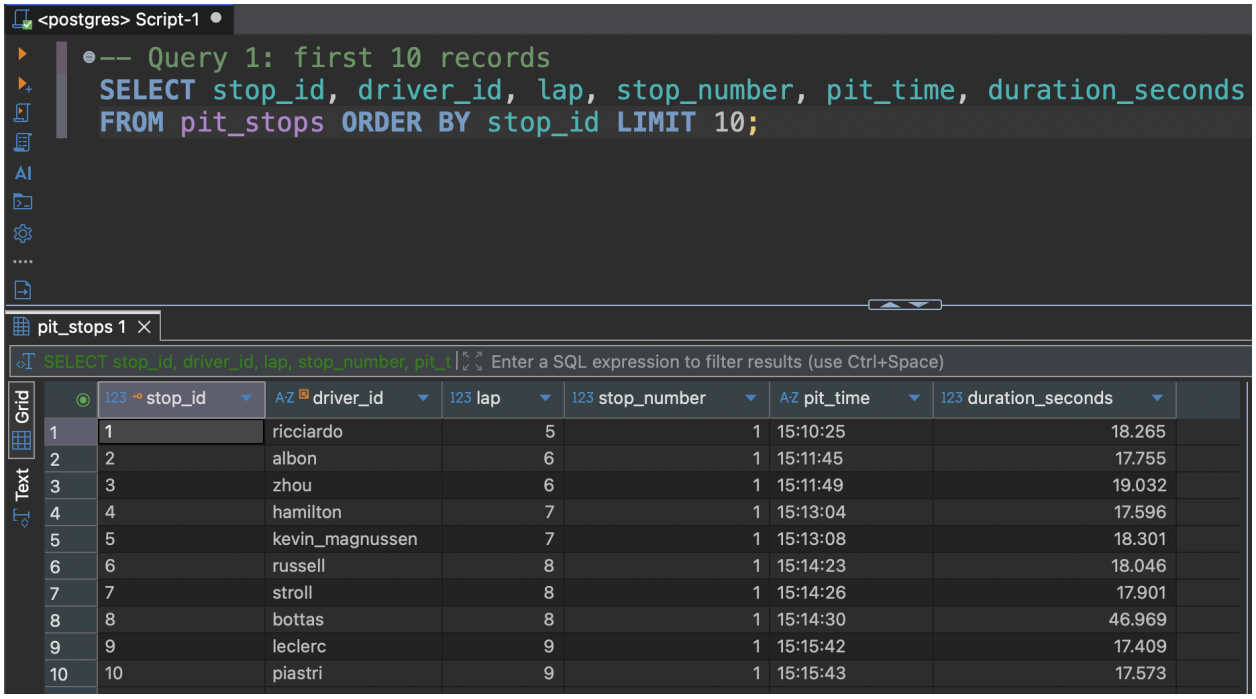
CREATE TABLE race (
    race_id INTEGER PRIMARY KEY,
    season INTEGER NOT NULL,
    round_number INTEGER NOT NULL,
    race_name TEXT NOT NULL,
    circuit_name TEXT NOT NULL,
    source_url TEXT NOT NULL,
    fetched_at TEXT NOT NULL
);

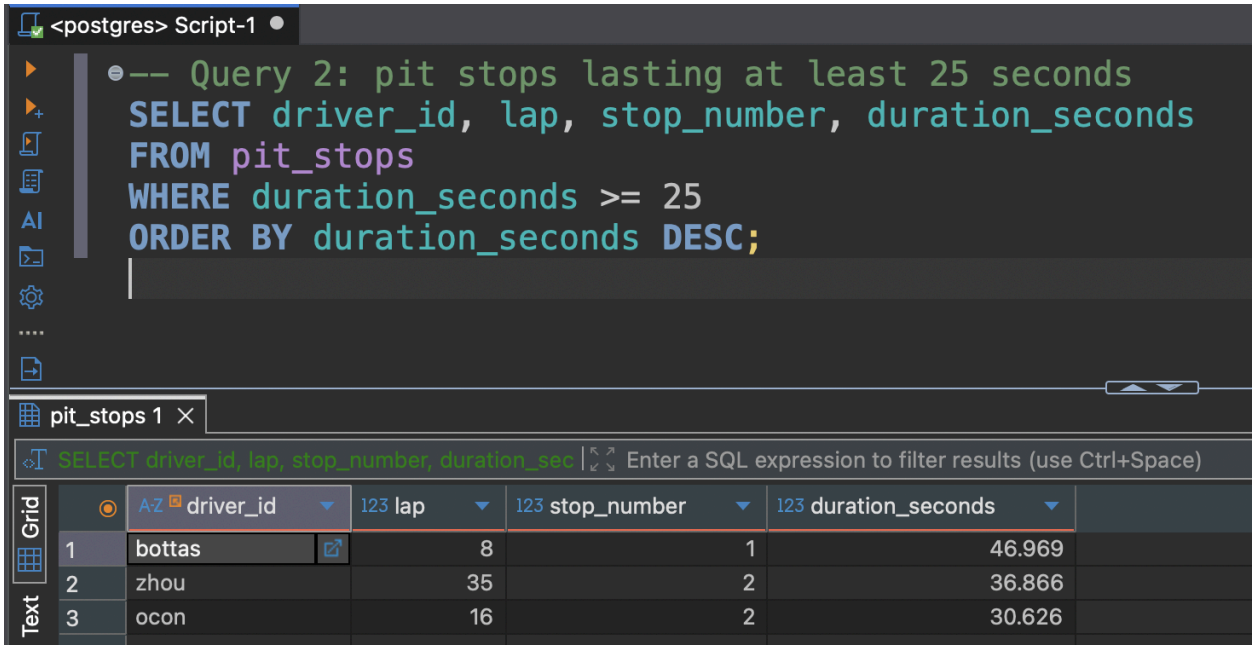
CREATE TABLE drivers (
    driver_id TEXT PRIMARY KEY,
    driver_name TEXT NOT NULL
);

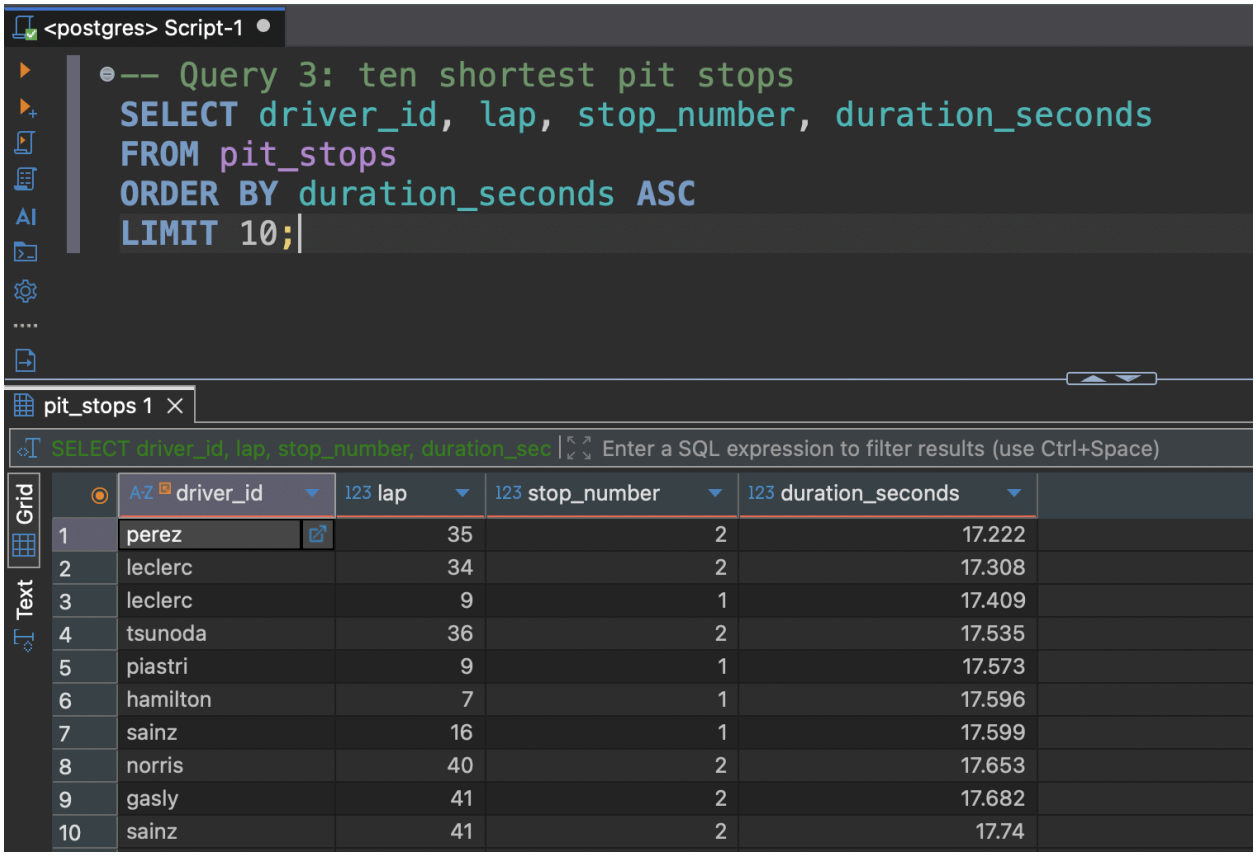
CREATE TABLE pit_stops (
    stop_id INTEGER PRIMARY KEY,
    driver_id TEXT NOT NULL,
    lap INTEGER NOT NULL CHECK (lap > 0),
    stop_number INTEGER NOT NULL CHECK (stop_number > 0),
    pit_time TEXT,
    duration_seconds REAL,
    FOREIGN KEY (driver_id) REFERENCES drivers(driver_id)
);

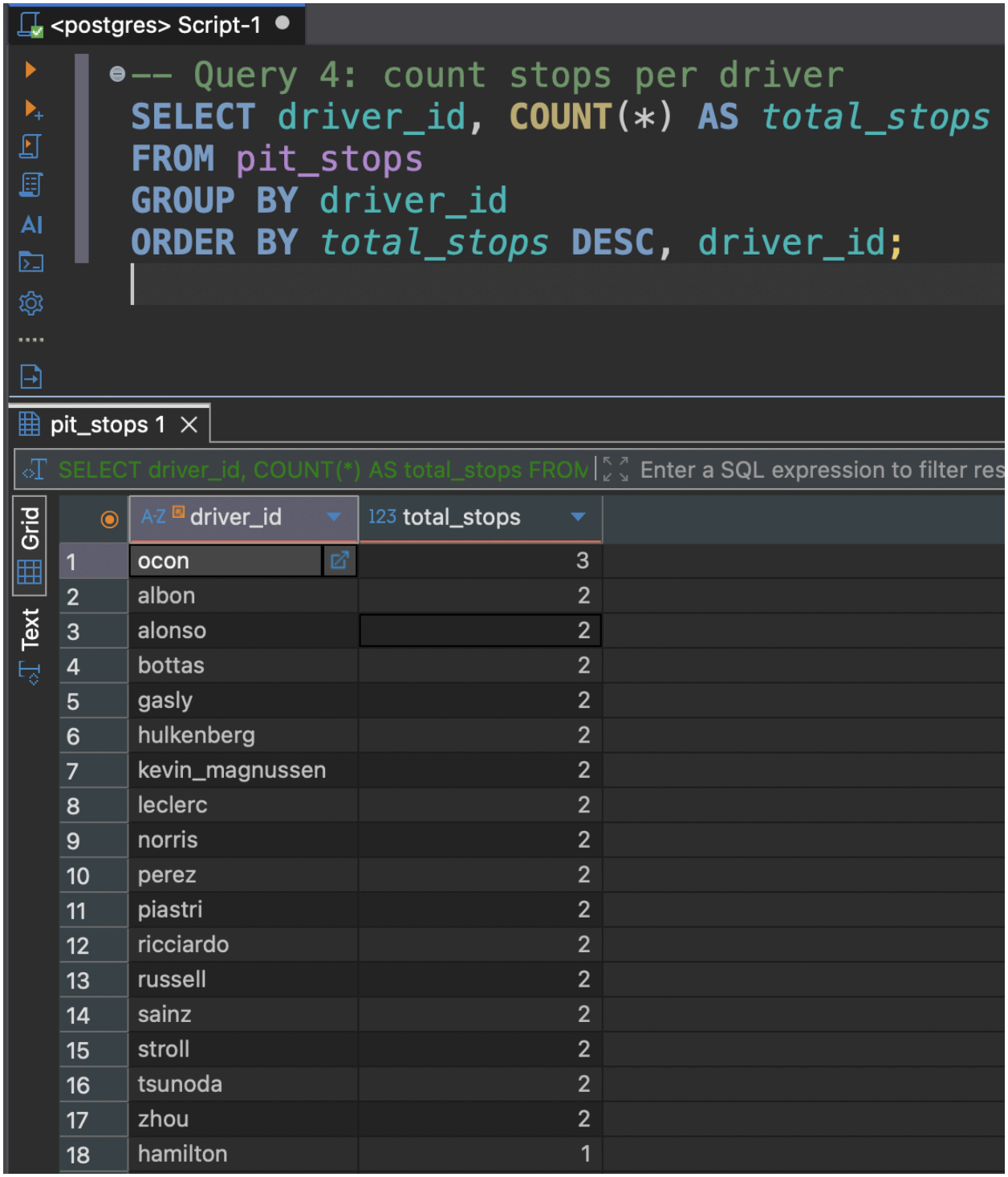
CREATE INDEX idx_pit_stops_driver ON pit_stops(driver_id);
CREATE INDEX idx_pit_stops_lap ON pit_stops(lap);
```

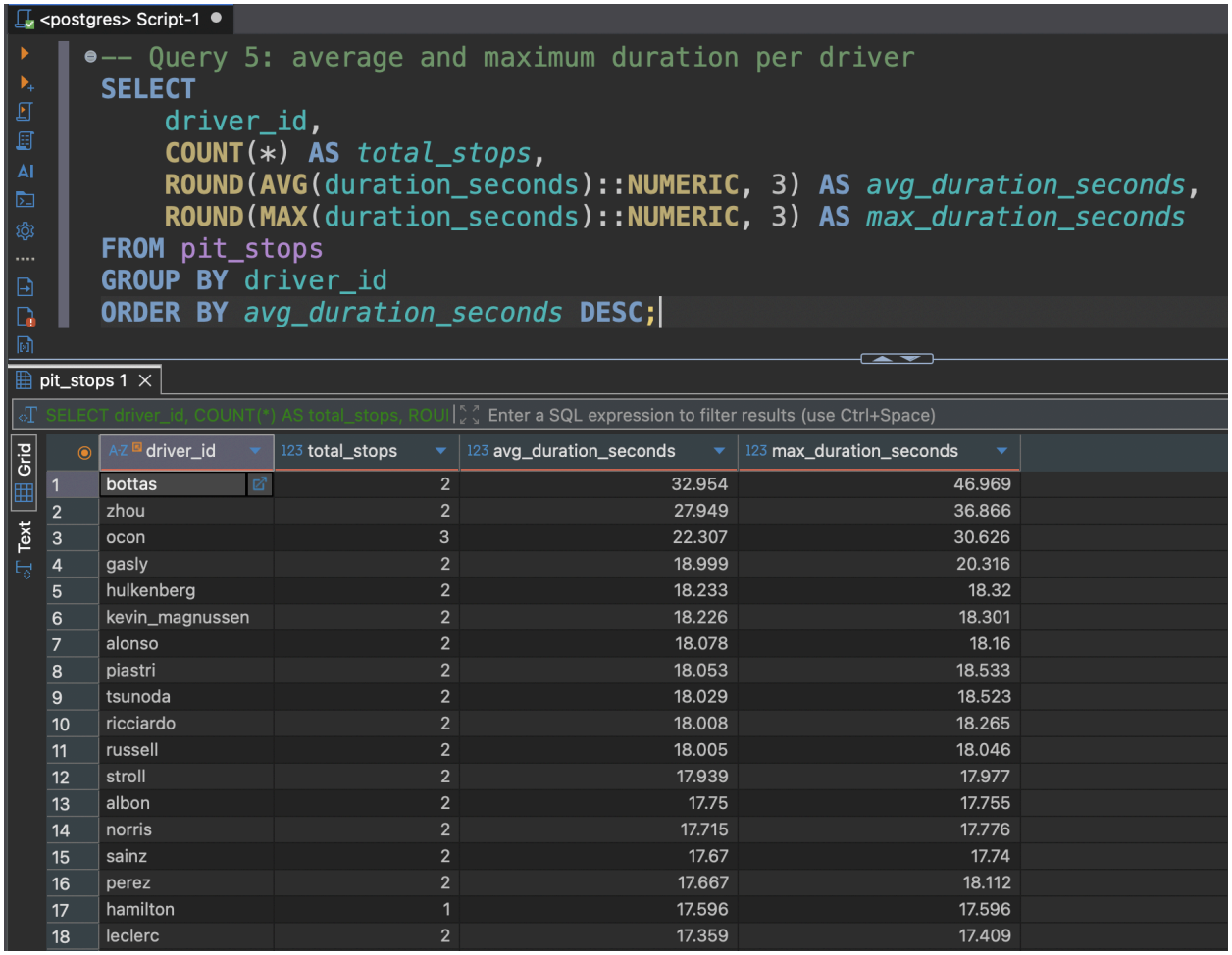

<postgres> Script-1
SELECT tablename
FROM pg_catalog.pg_tables
WHERE schemaname != 'pg_catalog'
AND schemaname != 'information_schema';
pg_tables 1
SELECT tablename FROM pg_catalog.pg_tables W
Enter a SQL expression to filter resu
Grid
Text
A-Z tablename
1 race
2 drivers
3 pit_stops

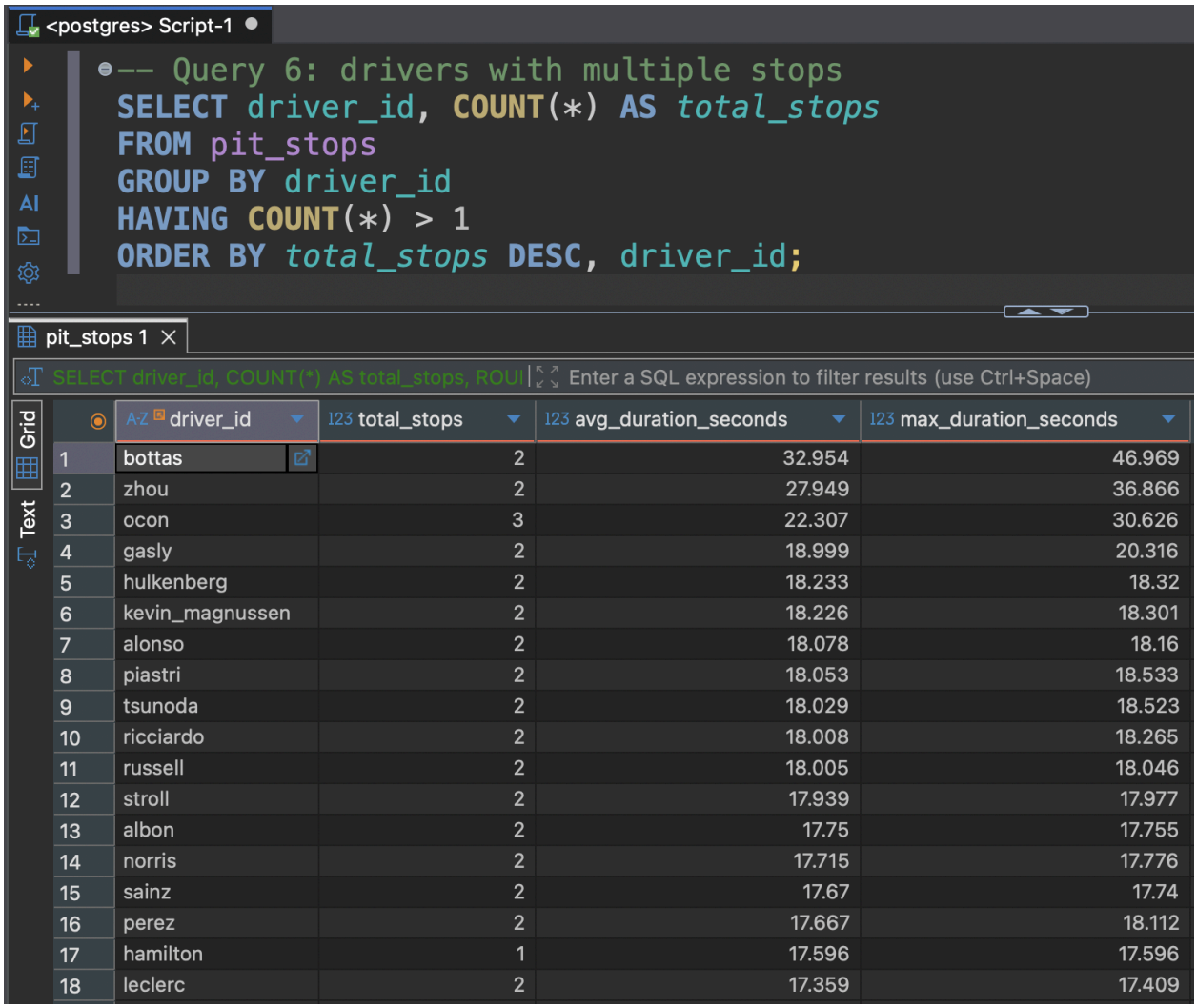

<postgres> Script-1
-- Query 1: first 10 records
SELECT stop_id, driver_id, lap, stop_number, pit_time, duration_seconds
FROM pit_stops ORDER BY stop_id LIMIT 10;
pit_stops 1
SELECT stop_id, driver_id, lap, stop_number, pit_t
Enter a SQL expression to filter results (use Ctrl+Space)
Grid
Text
stop_id driver_id lap stop_number pit_time duration_seconds
1 1 ricciardo 5 1 15:10:25 18.265
2 2 albon 6 1 15:11:45 17.755
3 3 zhou 6 1 15:11:49 19.032
4 4 hamilton 7 1 15:13:04 17.596
5 5 kevin_magnussen 7 1 15:13:08 18.301
6 6 russell 8 1 15:14:23 18.046
7 7 stroll 8 1 15:14:26 17.901
8 8 bottas 8 1 15:14:30 46.969
9 9 leclerc 9 1 15:15:42 17.409
10 10 piastri 9 1 15:15:43 17.573

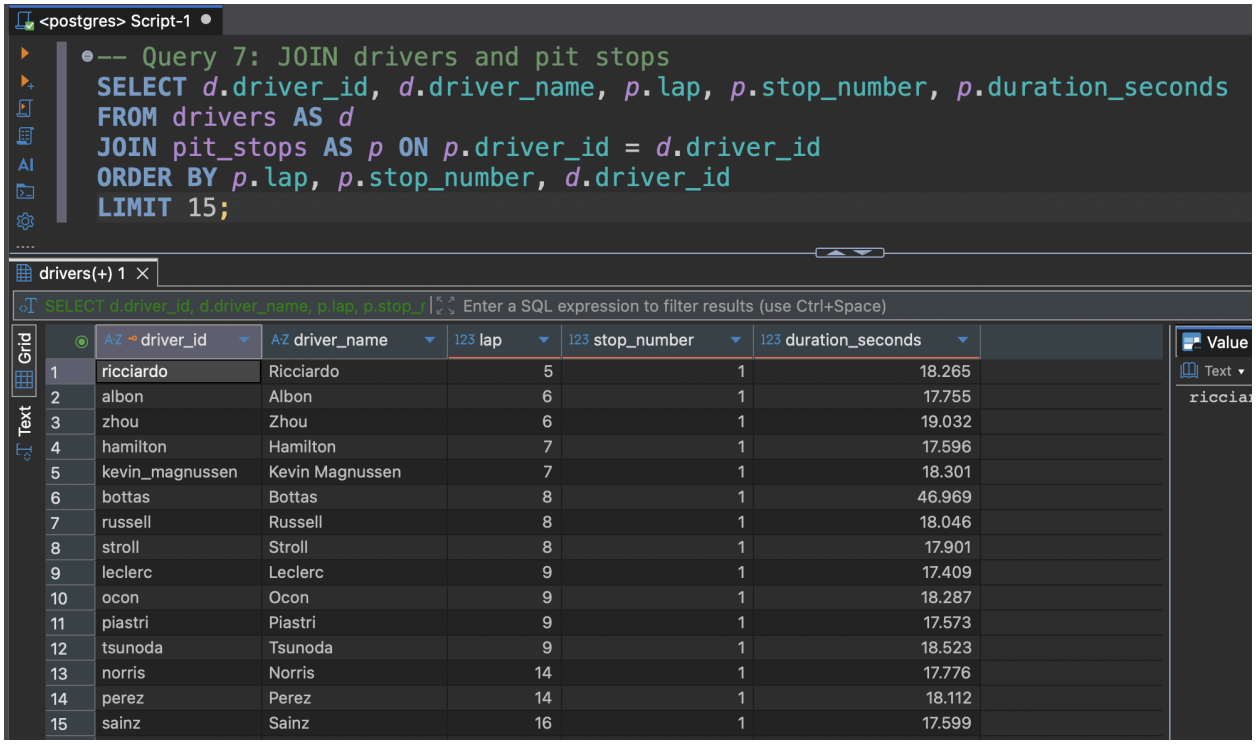

<postgres> Script-1
-- Query 2: pit stops lasting at least 25 seconds
SELECT driver_id, lap, stop_number, duration_seconds
FROM pit_stops
WHERE duration_seconds >= 25
ORDER BY duration_seconds DESC;
pit_stops 1
SELECT driver_id, lap, stop_number, duration_sec
Enter a SQL expression to filter results (use Ctrl+Space)
Grid
Text
driver_id lap stop_number duration_seconds
1 bottas 8 1 46.969
2 zhou 35 2 36.866
3 ocon 16 2 30.626

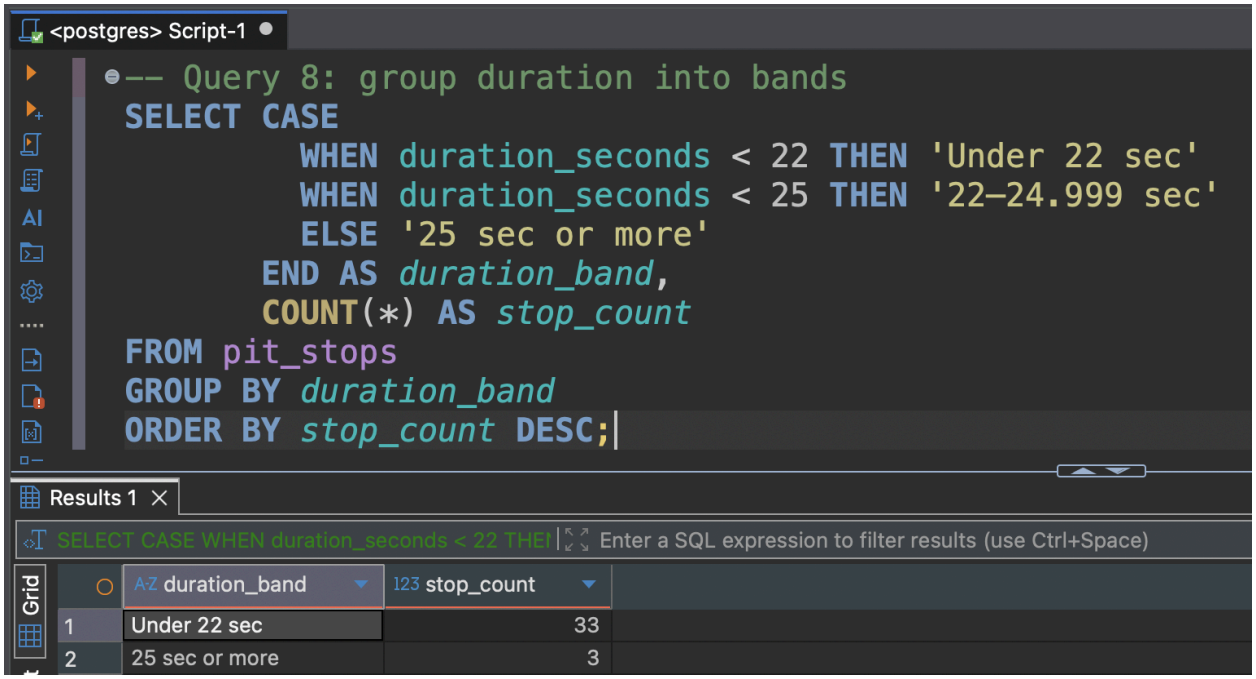

<postgres> Script-1
-- Query 3: ten shortest pit stops
SELECT driver_id, lap, stop_number, duration_seconds
FROM pit_stops
ORDER BY duration_seconds ASC
LIMIT 10;
pit_stops 1
SELECT driver_id, lap, stop_number, duration_sec
Enter a SQL expression to filter results (use Ctrl+Space)
Grid
Text
driver_id
lap
stop_number
duration_seconds
1 perez 35 2 17.222
2 leclerc 34 2 17.308
3 leclerc 9 1 17.409
4 tsunoda 36 2 17.535
5 piastri 9 1 17.573
6 hamilton 7 1 17.596
7 sainz 16 1 17.599
8 norris 40 2 17.653
9 gasly 41 2 17.682
10 sainz 41 2 17.74


<postgres> Script-1
-- Query 4: count stops per driver
SELECT driver_id, COUNT(*) AS total_stops
FROM pit_stops
GROUP BY driver_id
ORDER BY total_stops DESC, driver_id;
pit_stops 1
SELECT driver_id, COUNT(*) AS total_stops FROM
Enter a SQL expression to filter res
Grid
Text
driver_id
total_stops
1 ocon 3
2 albon 2
3 alonso 2
4 bottas 2
5 gasly 2
6 hulkenberg 2
7 kevin_magnussen 2
8 leclerc 2
9 norris 2
10 perez 2
11 piastri 2
12 ricciardo 2
13 russell 2
14 sainz 2
15 stroll 2
16 tsunoda 2
17 zhou 2
18 hamilton 1


<postgres> Script-1
-- Query 5: average and maximum duration per driver
SELECT
driver_id,
COUNT(*) AS total_stops,
ROUND(AVG(duration_seconds)::NUMERIC, 3) AS avg_duration_seconds,
ROUND(MAX(duration_seconds)::NUMERIC, 3) AS max_duration_seconds
FROM pit_stops
GROUP BY driver_id
ORDER BY avg_duration_seconds DESC;
pit_stops 1
SELECT driver_id, COUNT(*) AS total_stops, ROU
Enter a SQL expression to filter results (use Ctrl+Space)
Grid
Text
driver_id
total_stops
avg_duration_seconds
max_duration_seconds
1 bottas 2 32.954 46.969
2 zhou 2 27.949 36.866
3 ocon 3 22.307 30.626
4 gasly 2 18.999 20.316
5 hulkenberg 2 18.233 18.32
6 kevin_magnussen 2 18.226 18.301
7 alonso 2 18.078 18.16
8 piastri 2 18.053 18.533
9 tsunoda 2 18.029 18.523
10 ricciardo 2 18.008 18.265
11 russell 2 18.005 18.046
12 stroll 2 17.939 17.977
13 albon 2 17.75 17.755
14 norris 2 17.715 17.776
15 sainz 2 17.67 17.74
16 perez 2 17.667 18.112
17 hamilton 1 17.596 17.596
18 leclerc 2 17.359 17.409


<postgres> Script-1
-- Query 6: drivers with multiple stops
SELECT driver_id, COUNT(*) AS total_stops
FROM pit_stops
GROUP BY driver_id
HAVING COUNT(*) > 1
ORDER BY total_stops DESC, driver_id;
pit_stops 1
SELECT driver_id, COUNT(*) AS total_stops, ROUI
Enter a SQL expression to filter results (use Ctrl+Space)
Grid
Text
driver_id
total_stops
avg_duration_seconds
max_duration_seconds
1 bottas 2 32.954 46.969
2 zhou 2 27.949 36.866
3 ocon 3 22.307 30.626
4 gasly 2 18.999 20.316
5 hulkenberg 2 18.233 18.32
6 kevin_magnussen 2 18.226 18.301
7 alonso 2 18.078 18.16
8 piastri 2 18.053 18.533
9 tsunoda 2 18.029 18.523
10 ricciardo 2 18.008 18.265
11 russell 2 18.005 18.046
12 stroll 2 17.939 17.977
13 albon 2 17.75 17.755
14 norris 2 17.715 17.776
15 sainz 2 17.67 17.74
16 perez 2 17.667 18.112
17 hamilton 1 17.596 17.596
18 leclerc 2 17.359 17.409


<postgres> Script-1
-- Query 7: JOIN drivers and pit stops
SELECT d.driver_id, d.driver_name, p.lap, p.stop_number, p.duration_seconds
FROM drivers AS d
JOIN pit_stops AS p ON p.driver_id = d.driver_id
ORDER BY p.lap, p.stop_number, d.driver_id
LIMIT 15;
drivers(+) 1
SELECT d.driver_id, d.driver_name, p.lap, p.stop_r
Enter a SQL expression to filter results (use Ctrl+Space)
Grid
Text
driver_id
driver_name
lap
stop_number
duration_seconds
Value
Text
1 ricciardo Ricciardo 5 1 18.265
2 albon Albon 6 1 17.755
3 zhou Zhou 6 1 19.032
4 hamilton Hamilton 7 1 17.596
5 kevin_magnussen Kevin Magnussen 7 1 18.301
6 bottas Bottas 8 1 46.969
7 russell Russell 8 1 18.046
8 stroll Stroll 8 1 17.901
9 leclerc Leclerc 9 1 17.409
10 ocon Ocon 9 1 18.287
11 piastri Piastri 9 1 17.573
12 tsunoda Tsunoda 9 1 18.523
13 norris Norris 14 1 17.776
14 perez Perez 14 1 18.112
15 sainz Sainz 16 1 17.599


<postgres> Script-1
-- Query 8: group duration into bands
SELECT CASE
WHEN duration_seconds < 22 THEN 'Under 22 sec'
WHEN duration_seconds < 25 THEN '22–24.999 sec'
ELSE '25 sec or more'
END AS duration_band,
COUNT(*) AS stop_count
FROM pit_stops
GROUP BY duration_band
ORDER BY stop_count DESC;
Results 1
SELECT CASE WHEN duration_seconds < 22 THEI
Enter a SQL expression to filter results (use Ctrl+Space)
Grid
duration_band
stop_count
1 Under 22 sec 33
2 25 sec or more 3